INCH-POUND
MIL-DTL-24787/5A(SH)
24 September 2020
SUPERSEDING
MIL-F-24787/5(SH)
3 September 1993

DETAIL SPECIFICATION SHEET

FITTINGS, END, TYPE D (HOSE TO HOSE, 90-DEGREE ELBOW), REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

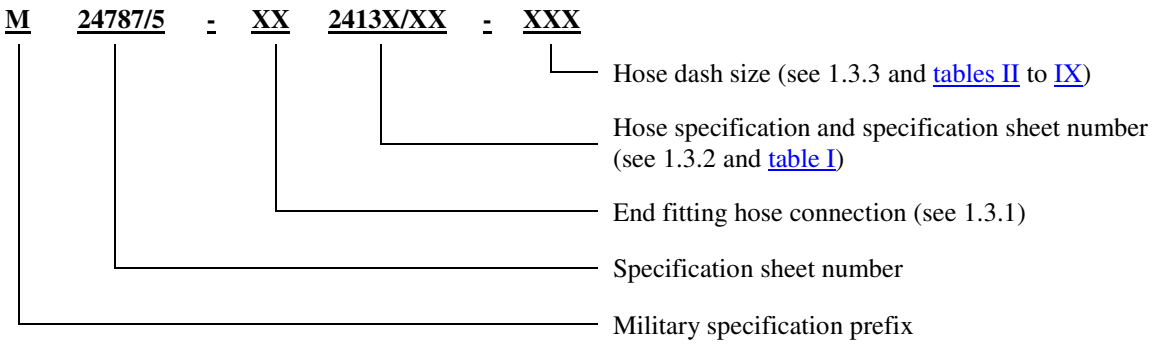
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24787.

1. SCOPE

1.1 Scope. This specification covers reusable hose to hose, 90-degree elbow end fittings (type D) for hose sizes ranging from -6 to -192 for use with wire- or synthetic fiber-reinforced rubber, flexible hose assemblies in accordance with MIL-DTL-24135 and MIL-DTL-24136.

1.2 Classification. The hose to hose, 90-degree elbow end fittings are type D as identified on figure 1. For the corresponding hose types, see table I.

1.3 Part or identifying number (PIN). The PIN is a definitive part number that is formed by combining the applicable specification sheet number (M24787/5 for MIL-DTL-24787/5), a dash, end fitting hose connection, a dash, hose specification and applicable specification sheet number, a dash, and hose dash size.



PIN example: M24787/5-RB24135/01-32X describes a MIL-DTL-24787/5 end fitting, with a ring and band hose connection, for use on MIL-DTL-24135/1, MIL-DTL-24135/5, and MIL-DTL-24135/13 hoses, with a hose size of -32.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3.1 End fitting hose connection. End fitting hose connections are identified in the PIN as follows:

Hose connection	PIN code
Solid Socket	SS
Ring and Band	RB
Bolt Together	BT

1.3.2 Hose specification and specification sheet number. Specification sheet numbers that only contain one digit will enter "0" as the first digit. For example, for specification sheet 1, the PIN code will be recorded as "01". For end fittings categorized with multiple specification sheets, the first specification and applicable specification sheet will be used for the PIN. For example, for an end fitting that can be used with MIL-DTL-24135/1, MIL-DTL-24135/5, and MIL-DTL-24135/13, the PIN code will be recorded as "24135/01".

1.3.3 Hose dash size. Hose dash sizes in tables II through IX that only contain one digit will enter "X" as the last digits. For example, for a hose dash size of -6, the PIN code will be recorded as "6XX". Hose dash sizes that contain two digits, will enter "X" as the last digit. For example, for a hose dash size of -12, the PIN code will be recorded as "12X". For hose dash sizes that contain three digits, all digits will be entered in the PIN. For example, for a hose dash size of -128, the PIN code will be recorded as "128".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-1368 - Tube and Pipe, Nickel-Copper Alloy, Seamless and Welded

MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded, (Copper Alloy Numbers 715 and 706)

MIL-DTL-24135 - Hose, Synthetic Rubber, Wire Reinforced, for Flexible Hose Assemblies, General Specification for

MIL-DTL-24135/1 - Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid or Double-Spiral and Single-Wire Braid), for Flexible Hose Assemblies (High Pressure, Sizes -4 through -32)

MIL-DTL-24135/2 - Hose, Synthetic Rubber, Wire Reinforced (Four-Ply Spiral-Wrapped or Double-Spiral and Single-Wire Braid), for Flexible Hose Assemblies (High Pressure, Sizes -6 through -16)

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MIL-DTL-24135/4	-	Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -40, -48, and -64)
MIL-DTL-24135/5	-	Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (Perforated Cover, for High Pressure Gas, Sizes -4 through -32)
MIL-DTL-24135/9	-	Hose, Synthetic Rubber, Wire Reinforced (Four-Ply, Heavy Spiral-Wrapped), for Flexible Hose Assemblies (High Pressure, Sizes -8 through -32)
MIL-DTL-24135/10	-	Hose, Synthetic Rubber, Wire Reinforced (Single- or Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -4 through -48)
MIL-DTL-24135/12	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Single-Wire Braid Over Single Synthetic Fiber Braid), for Flexible Hose Assemblies (Medium Pressure, High Temperature, Multi-Fluid, Sizes -4 through -40)
MIL-DTL-24135/13	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (High Pressure, High Temperature, Multi-Based Oil, Sizes -4 through -32)
MIL-DTL-24136	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies, General Specification for
MIL-DTL-24136/1	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -80 through -192)
MIL-DTL-24136/2	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (High Pressure, Sizes -4 through -32)
MIL-DTL-24136/3	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -4 through -32)
MIL-DTL-24136/4	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -40, -48, and -64)
MIL-C-24723	-	Castings, Nickel-Copper Alloy
MIL-DTL-24787	-	Fittings, End, Reusable, for Flexible Hose Assemblies, General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings
ASTM B271/B271M	-	Standard Specification for Copper-Based Alloy Centrifugal Castings
ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM B505/B505M	-	Standard Specification for Copper Alloy Continuous Castings

(Copies these documents are available online at www.astm.org.)

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SAE INTERNATIONAL

SAE AMS6381 - Steel, Mechanical Tubing 0.95Cr – 0.20Mo (0.38 – 0.43C) (SAE 4140)

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Hose to hose, 90-degree elbow end fittings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Hose to hose, 90-degree elbow end fittings. Hose to hose, 90-degree elbow end fittings shall be in accordance with the requirements of MIL-DTL-24787 and as specified herein.

3.2.1 Dimensions. End fitting dimensions shall be in accordance with figure 1 and tables II through IX (see 6.2). All dimensions in tables II through IX are recorded in inches.

3.2.2 Materials. End fitting materials shall be in accordance with figure 1 and tables II through IX. Nickel-based investment castings, if used, shall be in accordance with the material requirements of ASTM A494/A494M, with the exception of M-25S, and the welding and NDT requirements of MIL-C-24723. M-25S made in accordance with ASTM A494/A494M shall be in accordance with MIL-C-24723 for mechanical property, welding, and NDT requirements.

3.2.3 End fitting availability. Hose to hose, 90-degree elbow end fittings (type D) are available in the hoses and sizes indicated in table I (see 6.2).

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TABLE I. Hose to hose, 90-degree elbow end fitting by hose specification sheet and size.

End fitting size (hose dash number)	Hose specification sheet							
	24135/1, /5, and /13	24135/2	24135/4 and /10 (-40 and -48)	24135/9	24135/10 (-4 through -32), /12, and 24136/3	24136/1	24136/4	24136/2
-6	X	X	--	--	X	--	--	X
-8	X	X	--	X	X	--	--	X
-10	--	--	--	--	X	--	--	--
-12	X	X	--	X	X	--	--	X
-16	X	X	--	X	X	--	--	X
-20	X	--	--	X	X	--	--	X
-24	X	--	--	X	X	--	--	X
-32	X	--	--	X	X	--	--	X
-40	--	--	X	--	X	--	X	--
-48	--	--	X	--	--	--	X	--
-64	--	--	X	--	--	--	X	--
-80	--	--	--	--	--	X	--	--
-96	--	--	--	--	--	X	--	--
-128	--	--	--	--	--	X	--	--
-160	--	--	--	--	--	X	--	--
-192	--	--	--	--	--	X	--	--

NOTE: X denotes where hose to hose, 90-degree elbow end fittings are available by hose specification sheet and size.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-DTL-24787. Hose to hose, 90-degree elbow end fittings are not normally subjected to qualification testing since they do not attach to piping. They are qualified for a particular subgroup whenever an end fitting of that subgroup is submitted to and passes the requisite qualification testing.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-DTL-24787.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hose to hose, 90-degree elbow end fittings to this specification are intended for use in shipboard or shore-to-ship piping systems where pressures do not exceed the flange rated working pressure and temperatures do not exceed that of the applicable hose.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of end fittings.

c. Hose dash number (see 3.2.1).

d. End fitting type, class, hose specifications, and size (hose by pipe or tube) (see 3.2.3).

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24787 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

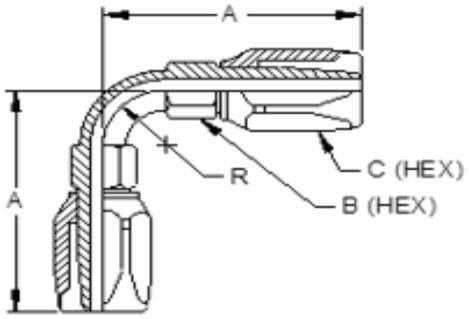
6.4 Subject term (key word) listing.

Synthetic fiber

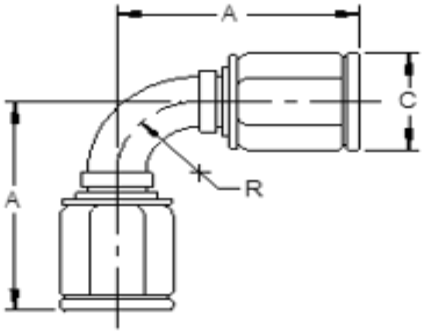
Wire

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

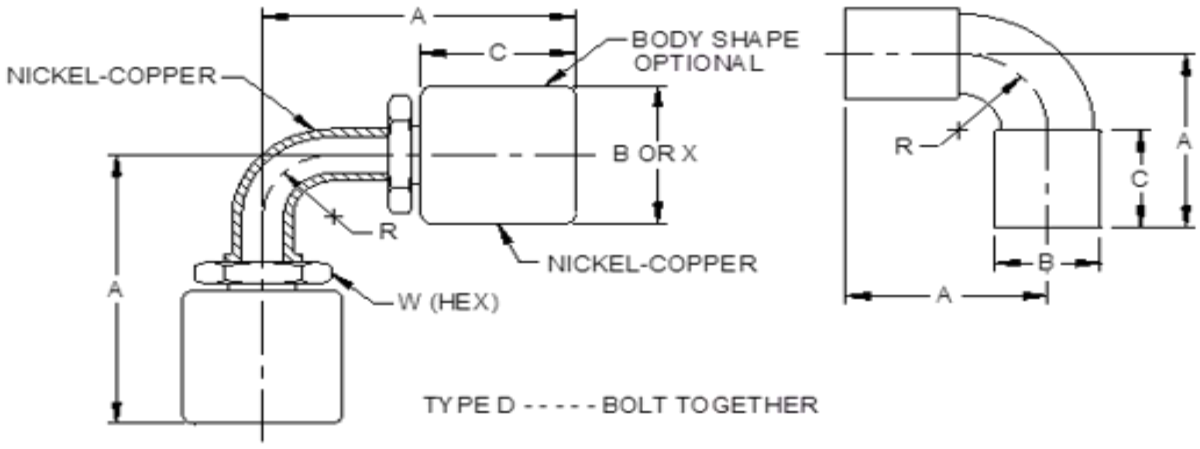
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TYPE D - - - - - SOLID SOCKET



TYPE D - - - - - RING AND BAND



TYPE D - - - - - BOLT TOGETHER

FIGURE 1. Type D end fittings.

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TABLE II. Dimensions for MIL-DTL-24135/1, /2, /5, and /13 solid socket 90-degree elbow end fittings.

Hose size (dash number)	A max	B nom	C max	R min
-6	3.55	0.69	1.25	0.50
-8	3.81	0.81	1.50	0.75
-12	4.75	1.06	1.75	1.25
-16	5.75	1.38	2.00	1.50
-20	6.75	1.62	2.75	1.75
-24	7.00	1.88	3.00	2.00
-32	8.00	2.50	3.50	2.50

NOTES:

1. Sizes -20 and larger do not have this hex.
2. The end fitting assembly materials shall be in accordance with QQ-N-281 or QQ-N-286.
3. The socket alternative material shall be in accordance with MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
4. The 90-degree tube elbow for sizes -6 through -12 shall be in accordance with MIL-T-1368 and for sizes -16 and larger, with MIL-T-16420.

TABLE III. Dimensions for MIL-DTL-24135/10 (-4 through -32 and -40), /12, and MIL-DTL-24136/3 solid socket 90-degree elbow end fittings.

Hose size (dash number)	A max	B nom	C max	R min
-6	2.45	0.56	0.81	0.50
-8	3.05	0.75	0.94	0.75
-10	3.57	0.81	1.12	1.00
-12	4.15	1.00	1.25	1.25
-16	3.93	1.38	1.44	1.50
-20	2.98	1/	1.75	1.22
-24	3.24	--	1.94	1.38
-32	3.89	--	2.44	1.62
-40	4.39	--	3.25	1.88

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TABLE III. Dimensions for MIL-DTL-24135/10 (-4 through -32 and -40), /12, and MIL-DTL-24136/3 solid socket 90-degree elbow end fittings – Continued.

FOOTNOTE:

1/ Sizes -20 and larger do not have this hex.

NOTES:

1. The end fitting assembly materials shall be in accordance with QQ-N-281 or QQ-N-286.
2. The socket alternative material shall be in accordance with MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
3. The 90-degree tube elbow for sizes -6 through -12 shall be in accordance with MIL-T-1368 and for sizes -16 and larger, with MIL-T-16420.

TABLE IV. Dimensions for MIL-DTL-24135/4 and /10 (-40 and -48 only) ring and band 90-degree elbow end fitting.

Hose size (dash number)	A max	C nom	R min
-40	5.70	3.90	1.88
-48	5.96	4.50	2.09
-64	6.75	5.44	2.81

NOTES:

1. Material for the nipple and socket assembly shall be UNS C95800 in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. The socket assembly may be multiple segment bolt or ring retained so long as envelope dimensions are not exceeded.
3. Socket assembly alternative material may be NiCu in accordance with QQ-N-281, QQ-N-286, MIL-C-24723, or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.

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TABLE V. Dimensions for MIL-DTL-24135/9 ring and band 90-degree elbow fitting.

Hose size (dash number)	A max	C max	R min
-8	4.45	1.60	0.75
-12	5.90	1.80	1.25
-16	6.40	2.16	1.50
-20	6.44	2.83	1.97
-24	7.25	3.16	2.00
-32	8.95	3.75	3.00

NOTES:

1. End fitting assembly materials for the hose nipple shall be in accordance with QQ-N-281 or QQ-N-286.
2. The socket for end fitting assemblies for -20 and larger may be one-piece or multiple-segment design, not exceeding the envelope dimensions. Sockets for sizes -8, -12, and -16 shall be one-piece design.
3. End fitting assembly materials for the socket shall be in accordance with QQ-N-281, QQ-N-286, MIL-C-24723, or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
3. Alternative material for the end fitting assembly socket for hoses size -20 and larger shall be steel conforming to AISI/SAE 4140 in accordance with SAE AMS6381, as applicable.
4. Tubing shall conform to MIL-T-16420.

TABLE VI. Dimensions for MIL-DTL-24135/4 and /10 (-40 and -48 only) bolt together 90-degree elbow end fitting.

Hose size (dash number)	A max	B nom	C nom	R min
-40	5.70	6.75	3.35	1.88
-48	5.96	7.75	3.35	2.09
-64	6.75	9.00	3.35	2.81

NOTES:

1. The hose nipple and socket assembly material shall be UNS C95800 in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. The socket assembly may be multiple-segment bolt or ring retained so long as envelope dimensions are not exceeded.
3. The socket assembly alternative material may be NiCu in accordance with QQ-N-281, QQ-N-286, or ASTM A494/A494M.

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TABLE VII. Dimensions for MIL-DTL-24136/1 bolt together 90-degree elbow end fittings.

Hose size (dash number)	A max	B nom	C nom	R min
-80	8.75	11.50	5.00	3.50
-96	9.25	12.00	5.00	4.00
-128	10.50	14.00	5.00	5.00
-160	12.00	16.50	5.00	6.38
-192	15.00	20.00	6.50	7.38

NOTES:

1. Material for segments and elbow end fittings shall be UNS C95800 bronze in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. Materials for fasteners shall be in accordance with QQ-N-281 or QQ-N-286.

TABLE VIII. Dimensions for MIL-DTL-24136/4 bolt together 90-degree elbow end fittings.

Hose size (dash number)	A max	B nom	C nom	R min
-40	6.25	6.75	5.00	2.13
-48	6.88	7.75	5.00	2.50
-64	7.00	9.00	5.00	3.00

NOTES:

1. Material for segment and elbow end fittings shall be UNS C95800 bronze in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. Material for fasteners shall be NiCu in accordance with QQ-N-281 or QQ-N-286.

TABLE IX. Dimensions for MIL-DTL-24136/2 bolt together 90-degree elbow end fittings.

Hose size (dash number)	A max	B nom	C nom	R min	W nom	X max
-6	3.25	1.50	2.00	0.50	0.94	2.47
-8	3.75	1.50	2.10	0.75	1.00	3.00
-12	4.75	2.25	2.42	1.25	1.31	3.50
-16	5.75	2.25	2.42	1.50	1.62	4.31
-20	6.75	2.75	2.80	1.75	2.00	4.81
-24	7.00	3.00	3.01	2.00	2.25	5.88
-32	8.00	3.50	3.88	2.50	2.75	5.81

NOTES:

1. End fitting assembly materials shall be in accordance with QQ-N-281 or QQ-N-286.
2. Material for elbow end fittings shall be in accordance with MIL-T-1368 or MIL-T-16420.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4730-2019-006)

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